

OCTOBER/NOVEMBER 2023 PAPER 1

1. $y = 2x + c$ — (1)

$y = x^2 + 3x + 1$ — (2)

(1) = (2): $2x + c = x^2 + 3x + 1$

$x^2 + x + 1 - c = 0$

Since the line is a tangent to the curve,

Discriminant = 0

$1^2 - 4(1)(1 - c) = 0$

$1 - 4 + 4c = 0$

$4c = 3$

$c = \frac{3}{4}$

EXAM TIP:

Since the line is a tangent to the curve, discriminant = 0.

2. Let $\frac{18 + 11x - 2x^2}{(x-1)(x+2)^2} = \frac{A}{x-1} + \frac{B}{(x+2)} + \frac{C}{(x+2)^2}$.

$18 + 11x - 2x^2 = A(x+2)^2 + B(x+2)(x-1) + C(x-1)$

Let $x = -2$: $-12 = -3C$

$C = 4$

Let $x = 1$: $27 = 9A$

$A = 3$

Comparing coefficients of x^2 ,

$-2 = 3 + B$

$B = -5$

$\therefore \frac{18 + 11x - 2x^2}{(x-1)(x+2)^2} = \frac{3}{x-1} - \frac{5}{(x+2)} + \frac{4}{(x+2)^2}$

EXAM TIP:Let $x = -2$ to eliminate the terms containing A and B . Then use substitution or equate coefficients to find the values of A and B .

3. LHS

$$= \frac{\cos^2 \theta}{(\operatorname{cosec} \theta - 1)(\operatorname{cosec} \theta + 1)} + \frac{\sin^2 \theta}{(\sec \theta - 1)(\sec \theta + 1)}$$

$$= \frac{\cos^2 \theta}{\operatorname{cosec}^2 \theta - 1} + \frac{\sin^2 \theta}{\sec^2 \theta - 1}$$

$$= \frac{\cos^2 \theta}{\cot^2 \theta} + \frac{\sin^2 \theta}{\tan^2 \theta}$$

$$= \frac{\cos^2 \theta}{\cos^2 \theta} + \frac{\sin^2 \theta}{\sin^2 \theta}$$

$$= \sin^2 \theta + \cos^2 \theta$$

$$= 1$$

$$= \text{RHS (proven)}$$

EXAM TIP:Apply the identities $\sin^2 \theta + \cos^2 \theta = 1$, $\tan^2 \theta + 1 = \sec^2 \theta$ and $1 + \cot^2 \theta = \operatorname{cosec}^2 \theta$.

4. (a) $\frac{d}{dx}(xe^{-2x}) = x(-2e^{-2x}) + e^{-2x}$
 $= e^{-2x} - 2xe^{-2x}$
 $= (1 - 2x)e^{-2x}$

(b) $\int (e^{-2x} - 2xe^{-2x}) dx = xe^{-2x} + c$

$\int e^{-2x} dx - \int 2xe^{-2x} dx = xe^{-2x} + c$

$\frac{e^{-2x}}{-2} - \int 2xe^{-2x} dx = xe^{-2x} + c$

$\int 2xe^{-2x} dx = -\frac{1}{2}e^{-2x} - xe^{-2x} + c$

$\int xe^{-2x} dx = -\frac{1}{4}e^{-2x} - \frac{1}{2}xe^{-2x} + c'$

EXAM TIP:From part (a), $\int (e^{-2x} - 2xe^{-2x}) dx = xe^{-2x} + c$. Then make $\int xe^{-2x} dx$ the subject.

5. $\frac{\cos \theta + 4 \sin \theta}{2 \cos \theta + \sin \theta} = \cot \theta$
 $= \frac{\cos \theta}{\sin \theta}$

$\sin \theta \cos \theta + 4 \sin^2 \theta = 2 \cos^2 \theta + \sin \theta \cos \theta$

$\tan^2 \theta = \frac{1}{2}$

$\tan \theta = \pm \frac{1}{\sqrt{2}}$

Basic angle = $0.615\,48$ (to 5 s.f.)

$\theta = -0.615, 0.615$ (to 3 s.f.)

EXAM TIP:Write $\frac{\sin^2 \theta}{\cos^2 \theta}$ as $\tan^2 \theta$. Calculate the values of θ in the first and fourth quadrants.

6. (a) $f(x) = \frac{ax^2}{x-a}$
 $f'(x) = \frac{(x-a)(2ax) - ax^2(1)}{(x-a)^2}$
 $= \frac{2ax^2 - 2a^2x - ax^2}{(x-a)^2}$
 $= \frac{ax^2 - 2a^2x}{(x-a)^2}$

(b) $g'(x) = (x-a)^2 f'(x)$

$$= (x-a)^2 \times \frac{ax^2 - 2a^2x}{(x-a)^2}$$

$$= ax^2 - 2a^2x$$

Since g is a decreasing function, $g'(x) < 0$.

$$ax^2 - 2a^2x < 0$$

$$ax(x-2a) < 0$$

Since $a > 0$, $x > 0$ and $x > a$,

$$a < x < 2a$$

By comparing, $2a = 8$

$$a = 4$$

EXAM TIP:

Since g is a decreasing function, then $g'(x) < 0$.

7. $kx^2 + 4x + k - 3 < 0$

Since the curve lies completely below the x -axis,

$$k < 0 \quad \text{and} \quad \text{Discriminant} < 0$$

$$4^2 - 4(k)(k-3) < 0$$

$$16 - 4k^2 + 12k < 0$$

$$4k^2 - 12k - 16 > 0$$

$$k^2 - 3k - 4 > 0$$

$$(k+1)(k-4) > 0$$

$$k < -1 \text{ or } k > 4$$

$$\therefore k < -1$$

EXAM TIP:

Since the curve lies entirely below the x -axis, discriminant < 0 .

8. $y - 2x = 12$ — (1)

$$x^2 - xy + y^2 = 63$$
 — (2)

From (1),

$$y = 2x + 12$$
 — (3)

Substitute (3) into (2):

$$x^2 - x(2x + 12) + (2x + 12)^2 = 63$$

$$x^2 - 2x^2 - 12x + 4x^2 + 48x + 144 = 63$$

$$3x^2 + 36x + 81 = 0$$

$$x^2 + 12x + 27 = 0$$

$$(x+3)(x+9) = 0$$

$$x = -3 \quad \text{or} \quad x = -9$$

$$y = 6 \quad \text{or} \quad y = -6$$

$\therefore$ Coordinates of these two points are $(-3, 6)$ and $(-9, -6)$

EXAM TIP:

Solve the two equations simultaneously to obtain the values of x .

9. (a) $y = 1 + x - \frac{x^2}{2} - 2x^3$

$$\frac{dy}{dx} = 1 - x - 6x^2$$

$$\text{When } \frac{dy}{dx} = 0,$$

$$1 - x - 6x^2 = 0$$

$$6x^2 + x - 1 = 0$$

$$(3x-1)(2x+1) = 0$$

$$x = \frac{1}{3} \quad \text{or} \quad x = -\frac{1}{2}$$

$\therefore$ x -coordinates are $\frac{1}{3}$ and $-\frac{1}{2}$

(b) $\frac{dy}{dx} = 1 - x - 6x^2$

$$\frac{d^2y}{dx^2} = -1 - 12x$$

For the gradient to be a maximum,

$$\frac{d^2y}{dx^2} = 0$$

$$-1 - 12x = 0$$

$$x = -\frac{1}{12}$$

$$\therefore x_P = -\frac{1}{12}$$

$$\begin{aligned} x\text{-coordinate of midpoint of } AB &= \frac{\frac{1}{3} + \left(-\frac{1}{2}\right)}{2} \\ &= -\frac{1}{12} \end{aligned}$$

$\therefore P$ and the midpoint of AB have the same

x -coordinate, $-\frac{1}{12}$. (shown)

EXAM TIP:

At the stationary points, $\frac{dy}{dx} = 0$. Since the gradient is a maximum, then $\frac{d^2y}{dx^2} < 0$.

10. (a) $x^2 + y^2 + 10x - 24y = 0$

$$(x+5)^2 - 25 + (y-12)^2 - 144 = 0$$

$$(x+5)^2 + (y-12)^2 = 169$$

Radius = 13 units

Coordinates of $C = (-5, 12)$

EXAM TIP:

Either complete the square to express it in the form $(x-a)^2 + (y-b)^2 = r^2$, where (a, b) are the coordinates of the centre and r is the radius, or apply the formulae for the general form of a circle.

(b) When $x = 0$,

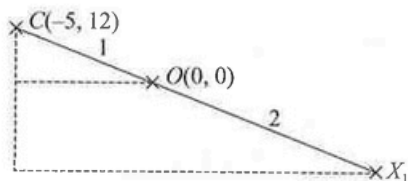
$$y^2 - 24y = 0$$

$$y(y-24) = 0$$

$$y = 0 \quad \text{or} \quad y = 24$$

$\therefore$ The circle intersects the y -axis at $(0, 0)$ and $(0, 24)$.

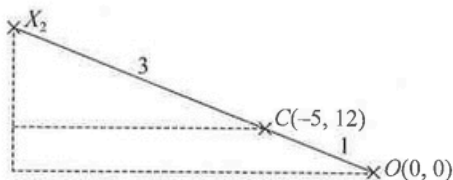
- (c) Case 1: O is between C and X



x -coordinate of $X_1 = 10$

y -coordinate of $X_1 = -24$

Case 2: C is between O and X .



x -coordinate of $X_2 = -20$

y -coordinate of $X_2 = 48$

$\therefore$ Possible positions of X are $(10, -24)$ and $(-20, 48)$

11. (a) $\frac{dr}{dt} = \frac{k}{2t+1}$

When $t = 0$, $\frac{dr}{dt} = 0.5$,

$$\frac{k}{2(0)+1} = 0.5$$

$$k = 0.5 \text{ (shown)}$$

(b) $\frac{dr}{dt} = \frac{0.5}{2t+1}$

$$r = \int \frac{0.5}{2t+1} dt$$

$$= 0.5 \int \frac{1}{2t+1} dt$$

$$= 0.25 \int \frac{2}{2t+1} dt$$

$$= 0.25 \ln(2t+1) + c$$

When $t = 0$, $r = 1$,

$$c = 1$$

$$\therefore r = 0.25 \ln(2t+1) + 1$$

(c) $A = \pi r^2$

$$\frac{dA}{dr} = 2\pi r$$

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$$

$$= 2\pi r \times \frac{0.5}{2t+1}$$

$$= 2\pi(0.25 \ln 7 + 1) \times \frac{0.5}{7}$$

$$= 0.667 \text{ cm}^2/\text{s} \text{ (to 3 s.f.)}$$

EXAM TIP:

Apply the equation involving the rates of change: $\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$.

12. (a) Initial height = 1.75 m

$$\begin{aligned} \text{(b)} \quad h &= 1.75 + 5t - 5t^2 \\ &= -5t^2 + 5t + 1.75 \\ &= -5(t^2 - t) + 1.75 \\ &= -5[(t - 0.5)^2 - 0.25] + 1.75 \\ &= -5(t - 0.5)^2 + 3 \\ &= 3 - 5(t - 0.5)^2 \end{aligned}$$

$$\therefore a = 3, b = -5, c = -0.5$$

- (c) Maximum height = 3 m, at time 0.5 s

EXAM TIP:

The maximum point of the graph of $y = a + b(t+c)^2$ is $(-c, a)$.

- (d) The ball was not thrown from ground level.

- (e) When $h \geq 2$,

$$3 - 5(t - 0.5)^2 \geq 2$$

$$5(t - 0.5)^2 \leq 1$$

$$(t - 0.5)^2 \leq 0.2$$

$$-\sqrt{0.2} \leq t - 0.5 \leq \sqrt{0.2}$$

$$-\sqrt{0.2} + 0.5 \leq t \leq \sqrt{0.2} + 0.5$$

$$\therefore \text{Length of time} = (\sqrt{0.2} + 0.5) - (-\sqrt{0.2} + 0.5)$$

$$= 2\sqrt{0.2}$$

$$= 0.894 \text{ s (to 3 s.f.)}$$

EXAM TIP:

A common mistake is simplifying $(t - 0.5)^2 \leq 0.2$ as $t - 0.5 \leq \pm\sqrt{0.2}$. This is incorrect.

13. (a) Gradient of $AB = \frac{h - (-7)}{2 - (-8)}$

$$= \frac{h + 7}{10}$$

Gradient of perpendicular bisector of AB

$$= -\frac{10}{h + 7}$$

(b) Midpoint of $AB = \left(\frac{2 + (-8)}{2}, \frac{h + (-7)}{2} \right)$

$$= \left(-3, \frac{h - 7}{2} \right)$$

Gradient of perpendicular bisector of AB

$$\frac{h - 7}{2} - \left(-\frac{13}{2} \right)$$

$$= \frac{h - 7}{2} + \frac{13}{2}$$

$$= \frac{h - 7}{2} + \frac{13}{2}$$

$$= \frac{h + 6}{2}$$

$$= \frac{h + 6}{2}$$

$$= \frac{h + 6}{2}$$

$$= \frac{h + 6}{2}$$

$$= \frac{h + 6}{2}$$

$$\begin{aligned}
 60 + 20h &= (h + 6)(h + 7) \\
 &= h^2 + 13h + 42 \\
 h^2 - 7h - 18 &= 0 \\
 (h - 9)(h + 2) &= 0 \\
 h &= 9 \text{ or } h = -2
 \end{aligned}$$

EXAM TIP:

Alternatively, use the midpoint of AB to find the equation of the perpendicular bisector of AB , and substitute the coordinates of X to find the values of h .

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$$\begin{aligned}
 1. \quad \left(2 - \frac{1}{2}x\right)^6 &= 2^6 - \binom{6}{1}(2^5)\left(\frac{1}{2}x\right) \\
 &\quad + \binom{6}{2}(2^4)\left(\frac{1}{2}x\right)^2 \\
 &\quad - \binom{6}{3}(2^3)\left(\frac{1}{2}x\right)^3 + \dots \\
 &= 64 - 96x + 60x^2 - 20x^3 + \dots \\
 (k + 2x)\left(2 - \frac{1}{2}x\right)^6 & \\
 &= (k + 2x)(64 - 96x + 60x^2 - 20x^3 + \dots) \\
 \text{Comparing coefficients of } x^3, & \\
 -20k + 120 &= 0 \\
 k &= 6
 \end{aligned}$$

EXAM TIP:

Apply the Binomial Theorem for the expansion of $\left(2 - \frac{1}{2}x\right)^6$ up to the term in x^3 . Find the terms in the expansion of $(k + 2x)\left(2 - \frac{1}{2}x\right)^6$ that yield the term in x^3 , and equate it to 0.

2. (a) $\angle ADB = \angle ABF$ ($\angle$ s in alt. segments)
 $\angle BAD = \angle FAB$ (common $\angle$)
 $\therefore \triangle ABD$ is similar to $\triangle AFB$. (Angle-Angle Similarity Test)
- (b) $\angle FBD = 90^\circ$ (rt. $\angle$ in a semicircle)
 Since BC is a diameter of a circle drawn through B, C and X , then $\angle BXC = 90^\circ$ (rt. $\angle$ in semicircle).
 $\angle FBX + \angle BXC = 90^\circ + 90^\circ$
 $= 180^\circ$
 By the converse of interior angles, $BF \parallel EC$ and $EBFC$ is a trapezium.
3. (a) When $t = 0$, $T_c = 86$.
 $\therefore$ Initial temperature $= 86^\circ\text{C}$

(b) $T_c = 86e^{-0.06t}$

When $T_c = 37$,

$$86e^{-0.06t} = 37$$

$$e^{-0.06t} = \frac{37}{86}$$

$$-0.06t = \ln \frac{37}{86}$$

$$t = -\frac{1}{0.06} \ln \frac{37}{86}$$

$$= 14.1 \text{ (to 3 s.f.)}$$

$\therefore$ It takes about 14.1 min for the temperature to drop to 37°C .

(c) (i) $T_f = 86e^{-\lambda t}$

When $t = 60$, $T_f = 82$,

$$82 = 86e^{-60\lambda}$$

$$e^{-60\lambda} = \frac{82}{86}$$

$$-60\lambda = \ln \frac{82}{86}$$

$$\lambda = -\frac{1}{60} \ln \frac{82}{86}$$

$$= 0.000794 \text{ (to 3 s.f.)}$$

(ii) When $T_c = \frac{1}{2}T_f$

$$86e^{-0.06t} = \frac{1}{2} (86e^{-0.00079380t})$$

$$\frac{e^{-0.06t}}{e^{-0.00079380t}} = \frac{1}{2}$$

$$e^{-0.059206t} = \frac{1}{2}$$

$$-0.059206t = \ln \frac{1}{2}$$

$$t = -\frac{1}{-0.059206} \ln \frac{1}{2}$$

$$= 11.7 \text{ (to 3 s.f.)}$$

$\therefore$ It takes about 11.7 min for the temperature of the tea in the cup to be half the temperature of the tea in the flask.

EXAM TIP:

Rearrange the terms so that the terms containing e are on the LHS of the equation.

$$\begin{aligned}
 4. \quad (a) \quad \tan 30^\circ &= \frac{\sqrt{3}}{3} \\
 \frac{2 \tan 15^\circ}{1 - \tan^2 15^\circ} &= \frac{\sqrt{3}}{3} \\
 6 \tan 15^\circ &= \sqrt{3} - \sqrt{3} \tan^2 15^\circ \\
 \sqrt{3} \tan^2 15^\circ + 6 \tan 15^\circ - \sqrt{3} &= 0 \\
 \tan 15^\circ &= \frac{-6 \pm \sqrt{6^2 - 4(\sqrt{3})(-\sqrt{3})}}{2(\sqrt{3})} \\
 &= \frac{-6 \pm \sqrt{48}}{2\sqrt{3}} \\
 &= \frac{-6 \pm 4\sqrt{3}}{2\sqrt{3}} \\
 &= -\frac{3}{\sqrt{3}} \pm 2 \\
 &= -\sqrt{3} \pm 2
 \end{aligned}$$

Since $\tan 15^\circ > 0$, then $\tan 15^\circ = 2 - \sqrt{3}$. (shown)

EXAM TIP:

Apply the Double Angle Formula: $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$. Then apply the Quadratic Formula to obtain an expression for $\tan 15^\circ$.

$$\begin{aligned}
 (b) \quad \tan 15^\circ - \tan 105^\circ &= \tan 15^\circ - \tan (60^\circ + 45^\circ) \\
 &= \tan 15^\circ - \frac{\tan 60^\circ + \tan 45^\circ}{1 - \tan 60^\circ \tan 45^\circ} \\
 &= (2 - \sqrt{3}) - \frac{\sqrt{3} + 1}{1 - \sqrt{3}} \\
 &= (2 - \sqrt{3}) - \frac{\sqrt{3} + 1}{1 - \sqrt{3}} \times \frac{1 + \sqrt{3}}{1 + \sqrt{3}} \\
 &= (2 - \sqrt{3}) - \frac{3 + 2\sqrt{3} + 1}{1 - 3} \\
 &= (2 - \sqrt{3}) - \frac{4 + 2\sqrt{3}}{-2} \\
 &= 2 - \sqrt{3} + 2 + \sqrt{3} \\
 &= 4 \text{ (proven)}
 \end{aligned}$$

EXAM TIP:

Apply the Addition Formula: $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$.

$$\begin{aligned}
 5. \quad (a) \quad y &= 3 \cos 3x - 5 \sin 3x \\
 \frac{dy}{dx} &= -9 \sin 3x - 15 \cos 3x \\
 \frac{d^2y}{dx^2} &= -27 \cos 3x + 45 \sin 3x \\
 p \frac{d^2y}{dx^2} + q \frac{dy}{dx} + 14y &= 34 \sin 3x \\
 &= p(-27 \cos 3x + 45 \sin 3x) \\
 &\quad + q(-9 \sin 3x - 15 \cos 3x) \\
 &\quad + 14(3 \cos 3x - 5 \sin 3x) + 34 \sin 3x \\
 &= -27p \cos 3x + 45p \sin 3x - 9q \sin 3x \\
 &\quad - 15q \cos 3x + 42 \cos 3x - 70 \sin 3x \\
 &\quad + 34 \sin 3x \\
 &= (-27p - 15q + 42) \cos 3x \\
 &\quad + (45p - 9q - 70 + 34) \sin 3x \\
 &= (42 - 27p - 15q) \cos 3x \\
 &\quad + (45p - 9q - 36) \sin 3x \\
 \therefore B &= 45p - 9q - 36
 \end{aligned}$$

EXAM TIP:

Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$, and substitute the expressions into the given equation. Group like terms and compare coefficients to find expressions for A and B .

$$\begin{aligned}
 (b) \quad \text{Since } A = B = 0, \\
 42 - 27p - 15q &= 0 \quad \text{--- (1)} \\
 45p - 9q - 36 &= 0 \quad \text{--- (2)} \\
 \text{From (1),} \\
 14 - 9p - 5q &= 0 \\
 9p + 5q &= 14 \quad \text{--- (3)} \\
 \text{From (2),} \\
 5p - q - 4 &= 0 \\
 q &= 5p - 4 \quad \text{--- (4)} \\
 \text{Substitute (4) into (3):} \\
 9p + 5(5p - 4) &= 14 \\
 9p + 25p - 20 &= 14 \\
 34p &= 34 \\
 p &= 1 \\
 \text{Substitute } p = 1 \text{ into (4):} \\
 q &= 5(1) - 4 = 1 \\
 \therefore p &= 1, q = 1
 \end{aligned}$$

$$\begin{aligned}
 6. \quad (a) \quad \text{Let } f(x) &= 6x^3 - 7x^2 - x + 2. \\
 f\left(-\frac{1}{2}\right) &= 6\left(-\frac{1}{2}\right)^3 - 7\left(-\frac{1}{2}\right)^2 - \left(-\frac{1}{2}\right) + 2 \\
 &= 0
 \end{aligned}$$

$\therefore$ By the Factor Theorem, $2x + 1$ is a factor. (shown)

$$\begin{aligned}
 f(x) &= 6x^3 - 7x^2 - x + 2 \\
 &= (2x + 1)(3x^2 + Bx + 2)
 \end{aligned}$$

Comparing coefficients of x ,

$$-1 = 4 + B$$

$$B = -5$$

$$\begin{aligned}\therefore f(x) &= (2x + 1)(3x^2 - 5x + 2) \\ &= (2x + 1)(3x - 2)(x - 1)\end{aligned}$$

EXAM TIP:

If $f\left(-\frac{b}{a}\right) = 0$, then $ax + b$ is a factor of $f(x)$.

Since $2x + 1$ is a factor of $f(x)$, use long division or equate coefficients to find the quadratic factor of $f(x)$.

$$(b) \quad 6(4^y) + 2(2^y) = 7(2^y) + 1$$

$$6(2^{2y}) + 2(2^y) = 7(2^y) + 1$$

$$6(2^{2y}) + 2(2^y) - 7(2^y) - 1 = 0$$

$$6(2^y)^2 - 5(2^y) - 1 = 0$$

Comparing with $f(x)$,

$$2^y = -\frac{1}{2} \quad \text{or} \quad 2^y = \frac{2}{3} \quad \text{or} \quad 2^y = 1$$

$$\begin{aligned}(\text{rejected}) \quad \lg \frac{2}{3} & \quad y = 0 \\ y &= \frac{\lg \frac{2}{3}}{\lg 2}\end{aligned}$$

$$= -0.585 \text{ (to 3 s.f.)}$$

$$\therefore y = -0.585 \text{ or } y = 0$$

EXAM TIP:

Manipulate the given equation into the same form as the expression given in part (a).

$$(c) \quad 2^y = \frac{2}{3}$$

$$\log_2 2^y = \log_2 \frac{2}{3}$$

$$y = \log_2 \frac{2}{3}$$

$$= \log_2 2 - \log_2 3$$

$$= 1 - \log_2 3 \text{ (shown)}$$

$$\therefore a = 2, b = 3$$

EXAM TIP:

Apply the Quotient Law of Logarithms: $\log_a x - \log_a y = \log_a \frac{x}{y}$.

$$\begin{aligned}7. \quad (a) \text{ Period} &= \frac{9\pi}{2} - \frac{7\pi}{2} \\ &= \pi\end{aligned}$$

$$\frac{2\pi}{b} = \pi$$

$$b = 2$$

Substitute $x = \frac{7\pi}{2}$ and $y = 10$ into $y = a \cos 2x$:

$$10 = a \cos 2\left(\frac{7\pi}{2}\right)$$

$$= a \cos 7\pi$$

$$= -a$$

$$a = -10$$

$$\therefore a = -10, b = 2$$

EXAM TIP:

For the graph of $y = a \cos bx$, the period is $\frac{2\pi}{b}$.

$$(b) \quad (i) \quad v = 5 \sin 0.2t + c$$

$$-1 \leq \sin 0.2t \leq 1$$

$$-5 \leq 5 \sin 0.2t \leq 5$$

$$-5 + c \leq 5 \sin 0.2t + c \leq 5 + c$$

For the particle not to change its direction,
 $v \geq 0$.

$$-5 + c \geq 0$$

$$c \geq 5$$

$\therefore$ Smallest value of $c = 5$

$$(ii) \quad v = 5 \sin 0.2t + 8$$

Total distance travelled in the third second

$$= \int_2^3 (5 \sin 0.2t + 8) dt$$

$$= \left[\frac{5(-\cos 0.2t)}{0.2} + 8t \right]_2^3$$

$$= [-25 \cos 0.2t + 8t]_2^3$$

$$= (-25 \cos 0.6 + 24) - (-25 \cos 0.4 + 16)$$

$$= 10.4 \text{ m (to 3 s.f.)}$$

EXAM TIP:

The distance travelled in the third second is not the same as the distance travelled in the first 3 seconds.

Apply: distance travelled $= \int_2^3 v dt$.

$$8. \quad (a) \quad \angle ABE = 180^\circ - 2\theta \text{ (sum of a } \triangle)$$

$$\angle BEC = 180^\circ - 2\theta \text{ (adj. } \angle\text{s on a str. line)}$$

$$\text{Area enclosed} = 3 \times \frac{1}{2} (r)(r) \sin (180^\circ - 2\theta)$$

$$= \frac{3}{2} r^2 \sin 2\theta \text{ m}^2$$

EXAM TIP:

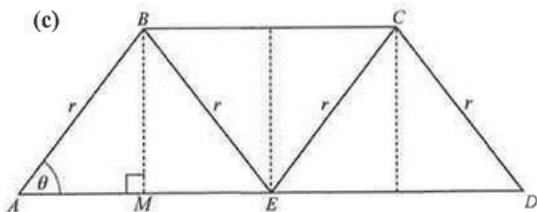
Area of a triangle $= \frac{1}{2} ab \sin C$.

$$(b) \quad \text{Maximum area} = \frac{3}{2} r^2 \text{ m}^2,$$

$$\text{when } \sin 2\theta = 1$$

$$2\theta = 90^\circ$$

$$\theta = 45^\circ$$



Consider $\triangle ABM$.

$$\cos \theta = \frac{AM}{r}$$

$$AM = r \cos \theta$$

Total length of wood = $4r + 6r \cos \theta$

$$8r \sin \theta = 4r + 6r \cos \theta$$

$$8r \sin \theta - 6r \cos \theta = 4r$$

$$2r(4 \sin \theta - 3 \cos \theta) = 4r$$

Since $r \neq 0$,

$$4 \sin \theta - 3 \cos \theta = 2 \text{ (shown)}$$

(d) Let $4 \sin \theta - 3 \cos \theta = R \sin(\theta - \alpha)$.

$$R = \sqrt{4^2 + 3^2}$$

$$= 5$$

$$\tan \alpha = \frac{3}{4}$$

$$\alpha = \tan^{-1} \frac{3}{4}$$

$$= 36.870^\circ \text{ (to 3 d.p.)}$$

$$\therefore 4 \sin \theta - 3 \cos \theta = 5 \sin(\theta - 36.870^\circ)$$

$$\text{When } 4 \sin \theta - 3 \cos \theta = 2,$$

$$5 \sin(\theta - 36.870^\circ) = 2$$

$$\sin(\theta - 36.870^\circ) = 0.4$$

$$\text{Basic angle} = 23.578^\circ \text{ (to 3 d.p.)}$$

$$\theta - 36.870^\circ = 23.578^\circ$$

$$\theta = 60.4^\circ \text{ (to 1 d.p.)}$$

EXAM TIP:

Apply the R-Formula: $a \sin \theta - b \cos \theta = R \sin(\theta - \alpha)$, where

$$R = \sqrt{a^2 + b^2} \text{ and } \tan \alpha = \frac{b}{a}.$$

9. (a) Consider $\triangle BCX$.

Using Pythagoras' Theorem,

$$XC^2 = 8^2 + (15 - x)^2$$

$$= 64 + 225 - 30x + x^2$$

$$= x^2 - 30x + 289$$

$$T = \frac{AX}{t_{AX}} + \frac{XC}{t_{XC}}$$

$$= \frac{x}{5} + \frac{\sqrt{x^2 - 30x + 289}}{3} \text{ (shown)}$$

(b) $T = \frac{x}{5} + \frac{\sqrt{x^2 - 30x + 289}}{3}$

$$\frac{dT}{dx} = \frac{1}{5} + \frac{1}{3} \times \frac{1}{2} (x^2 - 30x + 289)^{-\frac{1}{2}} (2x - 30)$$

$$= \frac{1}{5} + \frac{x - 15}{3\sqrt{x^2 - 30x + 289}}$$

When $\frac{dT}{dx} = 0$,

$$\frac{1}{5} + \frac{x - 15}{3\sqrt{x^2 - 30x + 289}} = 0$$

$$\frac{1}{5} = \frac{15 - x}{3\sqrt{x^2 - 30x + 289}}$$

$$3\sqrt{x^2 - 30x + 289} = 75 - 5x$$

$$9(x^2 - 30x + 289) = 5625 - 750x + 25x^2$$

$$9x^2 - 270x + 2601 = 5625 - 750x + 25x^2$$

$$16x^2 - 480x + 3024 = 0$$

$$x^2 - 30x + 189 = 0 \text{ (shown)}$$

EXAM TIP:

Apply the Chain Rule to find $\frac{dT}{dx}$. For a stationary value of T , $\frac{dT}{dx} = 0$.

(c) $x^2 - 30x + 189 = 0$

$$(x - 9)(x - 21) = 0$$

$$x = 9 \quad \text{or} \quad x = 21 \text{ (rejected)}$$

$$\text{Minimum time taken} = \frac{9}{5} + \frac{\sqrt{9^2 - 30(9) + 289}}{3}$$

$$= 5 \frac{2}{15} \text{ h}$$

$$= 5 \text{ h } 8 \text{ min}$$

EXAM TIP:

Reject $x = 21$ because $AX < 15 \text{ km}$.